



IMAGINE BUILDING DIFFERENTLY

PRODUCT DATA SHEET

59-Roof Assembly B.PUBLIC Prefab

PRODUCT OVERVIEW

Product Description

B.PUBLIC Prefab panelized assemblies exemplify best practices in construction and building science for unparalleled performance, durability, health, and environmental sustainability. The panels are manufactured in controlled environments leveraging both traditional craftsmanship and the latest technology to deliver airtight, vapor-open, and carbon-storing assemblies that achieve Net Zero and Passive House standards. The B.PUBLIC envelope provides industry leading comfort, quiet, and speed to new construction.

Application

The 59-Roof is designed to meet typical type V construction requirements up to 3 stories coast to coast and is suited for construction in IECC climate regions 3, through 7. B.PUBLIC assemblies and component connections minimize thermal bridging for unprecedented design freedom without sacrificing performance. B.PUBLIC Prefab roof assemblies are available in **Low-slope, Pitched, and Shed** configurations and custom options.



Installation

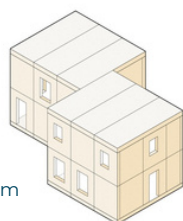
All prefabricated assemblies come fully wrapped in rated weather resistant barriers for durability during delivery and installation. Our unique isolated cavity design coupled with a redundant air-barrier ensures that prefabricated assemblies exceed Passive House level airtightness requirements.

Key Attributes

- Breathable yet water-proof vapor-open assembly
- Class A flame and smoke resistant materials
- Sealed to eliminate energy loss due to wind washing
- Grade 1 highest quality installation rating
- Industry leading acoustic performance
- Minimized thermal bridging per PHI standards
- Biogenic materials / Carbon sequestering / Low embodied energy
- > 70% post consumer recycled content
- Living Building Challenge compliant / Red List free
- Urea-Formaldehyde & foam free / No dangerous off-gassing

TECHNICAL OVERVIEW

Description	Off-site prefabricated light wood frame roof assembly
R-Value	16" Cellulose / R-3.7 per inch
Vapor Control	Class II / vapor variable membrane 0.13 to >13.0 perms Rated WRB- 2 months exposure
Cavity Insulation	Dense-pack Cellulose / Borate treated for fire / mold / insect resistance.
Weather Barrier	Fully adhered monolithic non-porous vapor-open weather barrier / rated 6 months exposure.
Thickness	16 5/8" (18 1/8" plus 1 1/2" furring for interior service cavity)



B.PUBLIC Prefab PBC
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B.PUBLIC is a woman-owned public benefit corporation. We're a team of architects, designers, tradespeople, and change makers meeting the climate and housing crises head-on. We're reimagining building for the better with technology that achieves extreme energy reduction, deep sustainability, and unmatched livability and comfort.





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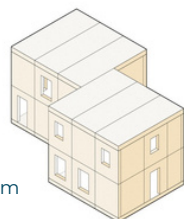
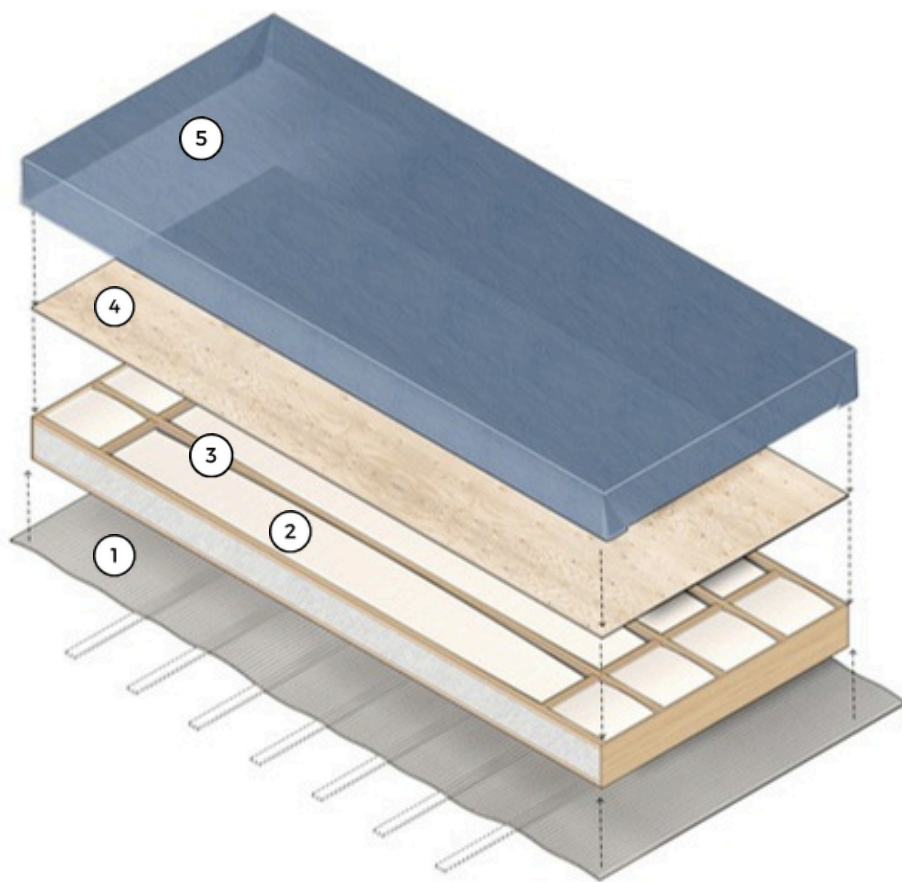
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MATERIAL SPECIFICATIONS

STANDARD ASSEMBLY SPECIFICATIONS*

Description	Off-site prefabricated light wood roof assembly
R-Value	16" Cellulose / R-3.7 per inch
Thickness	16 5/8" (18 1/8" plus 1 1/2" furring for interior service cavity)
①	Interior Membrane IBC / IRC Class II vapor retarder Vapor variable membrane 0.13 to >13.0 perms WRB -rated for 2 months exposure Class A Flame spread index 0 Class A Smoke developed index 35 ICC -ES report ESR-4854 Air permeance 0.00004 cfm/ft
②	Cavity Insulation Dense-pack Cellulose >3.5 lb per cubic ft. Borate treated for fire / mold / insect resistance. R-3.70 per inch 85% recycled content Formaldehyde free. -35 GWP value kg CO2e Class A fire rated
③	Structural Framing 16" APA web joist @ 24" OC with 1 1/8" engineered rim joist Continuous blocking
④	Structural Sheathing 5/8" APA rated sheathing Glued & nailed
⑤	Exterior Weather Barrier Fully adhered monolithic non-porous vapor-permeable weather resistant barrier. Rated 6 months exposure. Self-sealing around nail & screw penetrations. Waterproof to >32ft water column Vapor permanence 11 perms Airtightness 0.00009 cfm/ft2 Class A fire rated FS:10 SDI:15
Dimensional Tolerances	3/16" +/- Width & Length <1/4" out-of-plane <0.3% x Length / in-square



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*Subject to change per project engineering and architectural requirements.

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